

Vendor Selection for Enterprise AI Infrastructure

Choosing the right partners for your AI journey is mission-critical, with long-term implications for **cost, compliance, and innovation**.

A comprehensive selection process integrates technical evaluation, business alignment, and governance considerations.



Why Vendor Selection Matters



High-Stakes Investment

Multi-million dollar GPU spend demands careful consideration

Consequences of Poor Selection

Lock-in, compliance failures, and scaling limitations

Benefits of Strategic Partnership

Cost savings, enterprise-grade support, and innovation leverage

Key Evaluation Criteria

1

Performance

GPU/TPU availability, SLAs, and latency metrics

2

Cost Structure

On-demand vs. reserved vs. spot pricing models

3

Compliance

GDPR, HIPAA, SOC2, PCI certification readiness

4

Integration

Compatibility with existing enterprise IT stack

5

Support & Roadmap

Enterprise SLAs and co-innovation potential

Cloud Hyperscalers

AWS

Breadth of AI services, enterprise adoption, SageMaker ecosystem

GCP

Strong AI-first tooling with Vertex AI platform and TPU advantages

Azure

Enterprise integration excellence (Active Directory, compliance)

Pros

- Mature, battle-tested ecosystems
- Global infrastructure footprint
- Comprehensive security controls

Cons

- Premium pricing structure
- Risk of vendor lock-in
- Complex cost management



Specialized GPU/AI Clouds

Key Players

CoreWeave, Lambda Labs,
RunPod, Paperspace

Advantages

- Lower GPU cost structures
- Flexible configuration options
- Startup-friendly environments

Limitations

- Fewer compliance certifications
- Smaller global footprint
- Less mature enterprise tooling

Best suited for: R&D, burst compute needs, and cost-sensitive training workloads

On-Prem / Hybrid Vendors

Hardware Providers

NVIDIA DGX, HPE, Dell, Lenovo AI appliances

Software Infrastructure

VMware/Nutanix hybrid stacks

Pros

- Maximum control
- Compliance advantages
- Data residency certainty

Cons

- Significant upfront cost
- Slower scaling capacity
- Internal management overhead

Ideal for: regulated industries

(finance, healthcare, government)

SaaS & Model API Providers



Major Players

OpenAI, Anthropic, Cohere, HuggingFace Inference APIs

Pros

- Fastest path to MVP delivery
- Zero infrastructure management
- Access to cutting-edge models

Cons

- High per-call costs at scale
- Limited control and customization
- Intellectual property concerns

Enterprises typically leverage for rapid prototyping & augmenting internal models



Risk & Lock-In Assessment



Proprietary APIs

Significant migration risk when deeply integrated



Regional Presence

Data residency and sovereignty considerations

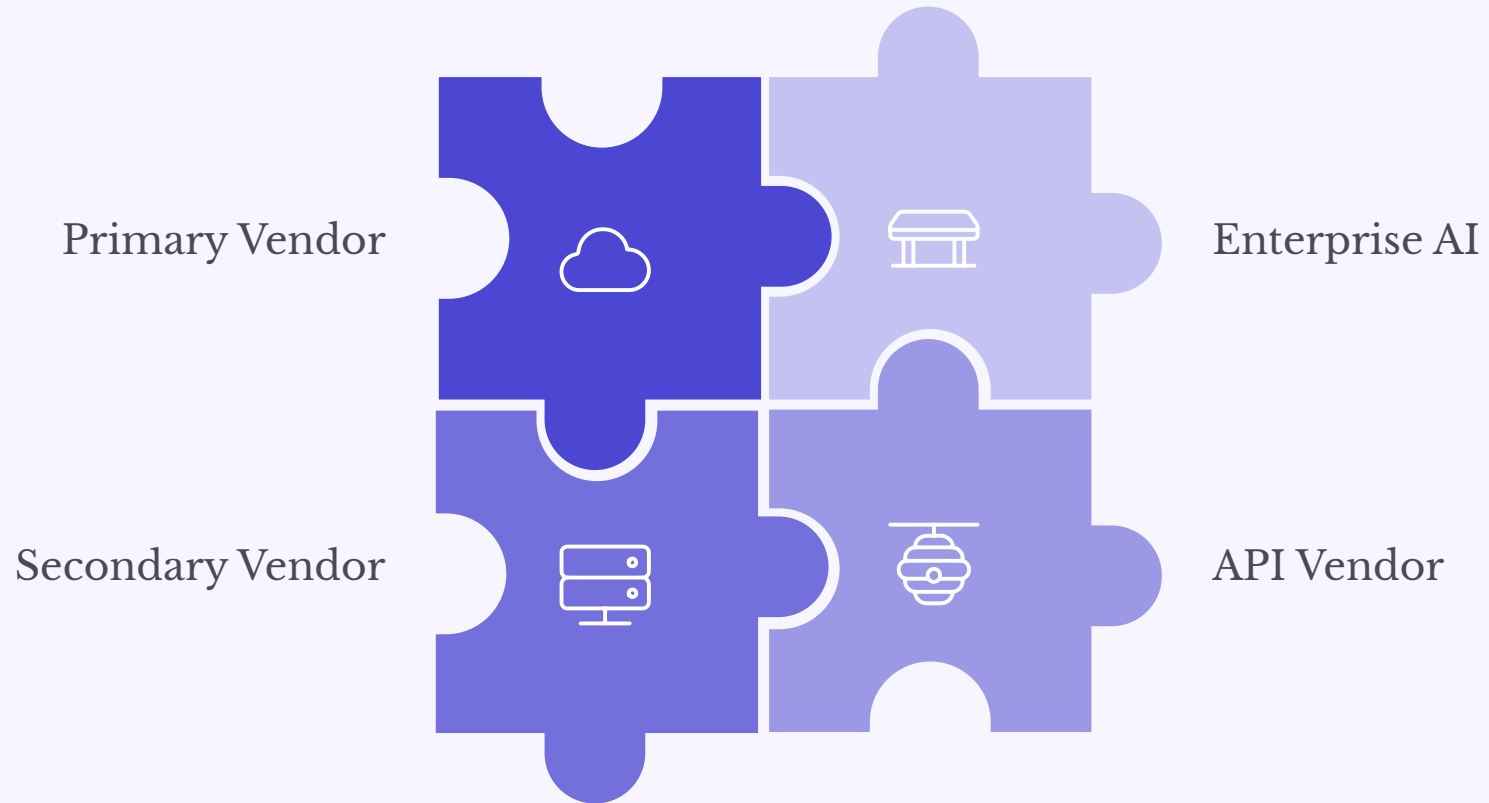


Licensing Terms

Export controls and hidden cost accelerators

Critical requirement: Evaluate your [exit strategy](#) before commitment

Multi-Vendor Strategy



A balanced multi-vendor approach provides **innovation speed** + **operational resilience** + cost optimization

Compliance & Security Checklist

Industry Certifications

SOC2, HIPAA, FedRAMP, ISO 27001

Data Governance

Regional residency guarantees and data handling practices

Encryption Controls

Customer-controlled encryption keys (BYOK/HYOK)

Identity Integration

RBAC + IAM integration with enterprise identity provider



Cost & FinOps Considerations



Capacity Planning

Reserved capacity vs. spot/preemptible instances based on workload predictability



Cost Attribution

Per-tenant chargeback/showback mechanisms for departmental accountability



Hardware Selection

Strategic GPU/TPU selection (A100 vs. L40 vs. H100) aligned with specific workload needs



Negotiation Leverage

Contract terms emphasizing portability and multi-cloud readiness to maintain flexibility



Case Study: Global Enterprise Vendor Mix

Core Infrastructure

Azure selected for seamless integration with existing Microsoft stack

R&D Workloads

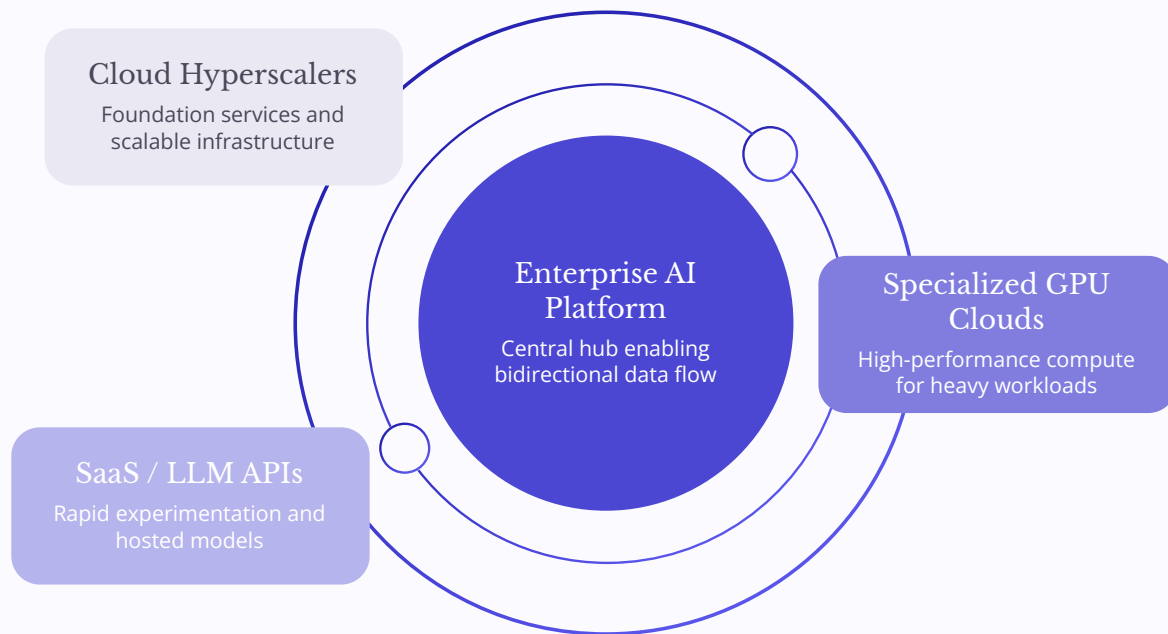
CoreWeave deployed for cost-effective GPU resources and specialized configurations

API Services

OpenAI utilized for rapid POCs, with strategic migration path to in-house models

Result: Cost control + innovation speed + compliance coverage

Balanced Vendor Ecosystem



"Balanced vendor mix = resilience + flexibility + innovation"

Summary: Strategic Vendor Selection

Comprehensive Evaluation

Assess vendors across performance, cost, compliance, support, and integration dimensions

Diversified Strategy

No single vendor solves all enterprise AI infrastructure needs

Multi-Vendor Approach

Reduce lock-in risk while optimizing for specific workload requirements

Strategic Balance

Enterprises must prioritize both **short-term wins** and **long-term resilience**

